

Packet Tracer - WAN Concepts

Objectives

In this activity, you will investigate various types of WANs by exploring a topology that uses diverse connectivity technologies.

Describe different WAN connectivity options.

Background / Scenario

You will explore WAN technologies that are used to connect business and home users to data services.

Note: There is no scoring in this activity.

Instructions

Part 1: Investigate Consumer WAN Technologies for Home and Mobile Devices

Step 1: Explore Consumer WAN Technologies

In this step, you will explore three consumer WAN technologies and home networks.

a. Look at the two home networks.

Question: What are the WAN technologies in use?

Answer: The two home networks use cable and DSL. The smartphone uses a third technology, cellular (3G/4G) data.

b. Examine the connections used in the network topology by selecting the Connections icon (the orange lightning bolt) in the PT devices menu. Hover over the media icons to display their names in the white box at the bottom of the PT window.

Question: What media is used to connect the two home networks to the ISP? What devices in the home networks are directly connected to the ISP?

Answer: The Home Cable Network reaches the ISP over coaxial cable, and the device on the home side of that link is the coaxial splitter. The Home DSL Network reaches the ISP over a telephone cable, and the device on the home side of that link is the DSL modem.

c. Click the DSL modem and open the Physical tab.

Question: What ports are available on the device and what is connected to them?

Answer: The DSL modem has two ports. One port connects to the telephone line coming from the Telco. The other port connects to the home LAN over Ethernet.

Question: What is the purpose of the DSL modem?

Answer: It converts the signal carried on the telephone network into Ethernet so the devices in the home network can use it (and converts in the other direction for outgoing traffic).

Question: What is the type of connection between the ISP/Telco/Cable Company network and the Home Cable Network? Why is the splitter necessary?

Answer: The connection is coaxial cable. The splitter is needed because the same cable carries both TV (video) signals and internet data. The splitter separates them, sending the data to the cable modem and the video to the TV.

d. Look at the ports on the cable modem.

Question: What does the cable modem do? What connections does it have?

Answer: The cable modem converts the cable data signal into Ethernet. It has a coaxial connection coming in from the splitter and a UTP (Ethernet) connection going out toward the home wireless router.

Question: Which port does the cable from the cable modem connect to on the home wireless router? Where did the interface IP address come from?

Answer: It plugs into the router's Internet port. The IP address on that interface was assigned by the ISP's network through DHCP.

e. Look at the Smartphone.

Question: What is its IP address? Where did the IP address come from?

Answer: The smartphone's IP address on its cellular (3G/4G) interface is 198.51.100.101. It was assigned by the Telco (cellular) network using DHCP.

Question: What data service is the cellphone currently using (cellular data or Wi-Fi)?

Answer: It is currently using cellular data (3G/4G), not Wi-Fi.

Step 2: Explore the Business WAN

In this step you will explore the business WAN. The business is a retail tire store. It has a local headquarters where most of the business functions occur, and three stores that are connected to the business WAN.

a. Look at the Connections menu.

Question: What different types of connections do you see in use in the Business network?

Answer: Ethernet over copper, Ethernet over fiber, and serial connections.

b. Open the physical view for the StoreNet switch.

Question: What types of interfaces are present? You may need to zoom and scroll the view to see.

Answer: The switch has Gigabit Ethernet copper ports plus four modular slots. Three of those slots hold GLC-LH-SMD fiber SFP (Small Form-Factor Pluggable) modules, which let the switch connect to fiber optic Ethernet links.

Question: Which interfaces and media are used to connect the store networks to the Business Headquarters network? Why was this done?

Answer: The stores connect to the StoreNet switch through the fiber SFP interfaces, using Ethernet over fiber optic cable. Fiber was chosen because of the distance between the stores and headquarters. In real life another provider would supply this fiber service; the activity simplifies that.

Question: What type of WAN service is used to connect the Business Headquarters router to the ISP?

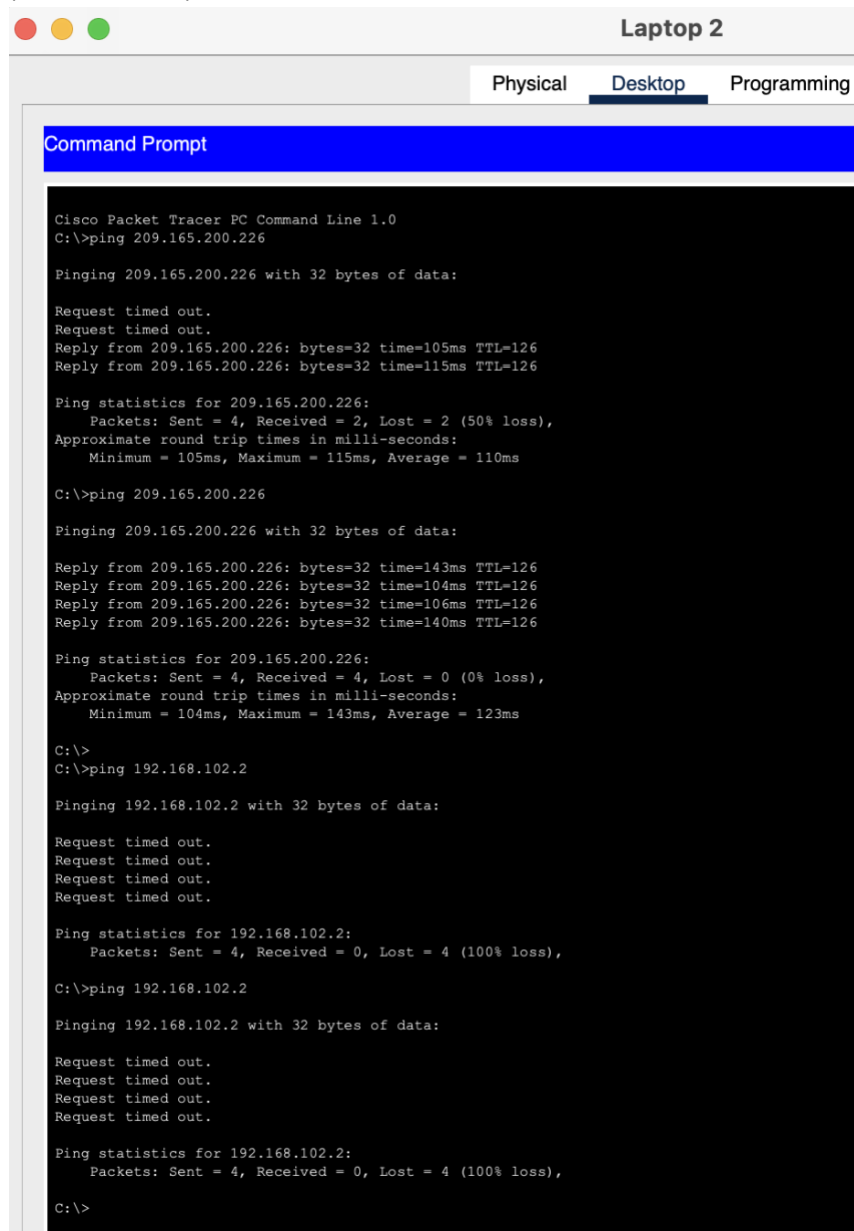
Answer: A serial WAN connection.

Part 2: Explore Connectivity

Question: Ping devices within the Business WAN and the Consumer WAN networks. Also ping between the networks and between the networks and the web server. Can all hosts ping each other and the web server?

Answer: No. In my tests, Laptop 2 (home network) and PC0 (Business network) could both ping the web server at 209.165.200.226. However, Laptop 2 could not ping PC0 (192.168.102.2); all four packets timed out (100% loss). So hosts in the Business network cannot reach hosts in the Consumer networks, and the reverse is also true, but both can reach the web server.

Figure 1: Laptop 2 (home network) pinging the web server (209.165.200.226) and PC0 (192.168.102.2).



The screenshot shows a laptop window titled "Laptop 2" with three tabs: "Physical", "Desktop", and "Programming". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the output of two ping commands. The first command is "C:\>ping 209.165.200.226", which shows successful results with 2 packets received and 50% loss. The second command is "C:\>ping 192.168.102.2", which shows 4 packets lost and 100% loss.

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Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 209.165.200.226: bytes=32 time=105ms TTL=126
Reply from 209.165.200.226: bytes=32 time=115ms TTL=126

Ping statistics for 209.165.200.226:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 105ms, Maximum = 115ms, Average = 110ms

C:\>ping 209.165.200.226

Pinging 209.165.200.226 with 32 bytes of data:

Reply from 209.165.200.226: bytes=32 time=143ms TTL=126
Reply from 209.165.200.226: bytes=32 time=104ms TTL=126
Reply from 209.165.200.226: bytes=32 time=106ms TTL=126
Reply from 209.165.200.226: bytes=32 time=140ms TTL=126

Ping statistics for 209.165.200.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 104ms, Maximum = 143ms, Average = 123ms

C:\>
C:\>ping 192.168.102.2

Pinging 192.168.102.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.102.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.102.2

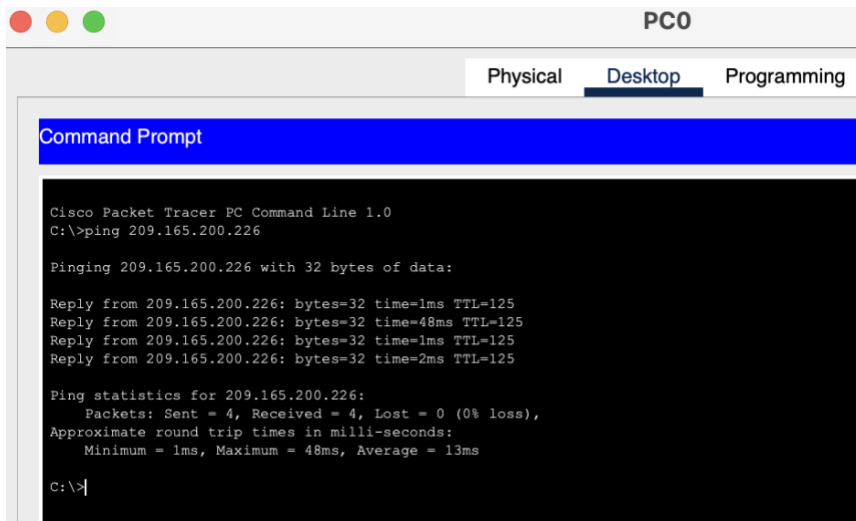
Pinging 192.168.102.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.102.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 2: PC0 (Business network) pinging the web server (209.165.200.226).



Question: Is this a good situation?

Answer: Yes. Business and home networks are private and should not be directly reachable from each other or from the outside, mainly for security reasons. Both can still reach the shared web server, which is what they need.